

10(a)

*any **eight** from:*

- 1 auxin binds to receptor ;
- 2 in cell surface membrane ;
- 3 (auxin) increases proton pump activity / described ;
- 4 (more) protons enter cell wall ;
- 5 cell wall, becomes more acidic / has reduced pH ;
- 6 expansins activated ;
- 7 (expansins) loosen / break, bonds between (cellulose) microfibrils ;
- 8 K^+ enters cell ;
- 9 water potential of cell decreases ;
- 10 more water can enter cell by osmosis / AW ;
- 11 turgor pressure / described ;
- 12 ref. to acid growth hypothesis ;

Question	Answer	Marks
10(b)	any seven from: 1 <i>idea that</i> phenotype results from interaction of genotype and environment ; 2 environment may, limit / modify, expression of gene(s) / AW ; 3 continuous variation example ; e.g. size / mass / height 4 qualified ; e.g. because, food / nutrients / ions, missing or malnutrition occurs 5 environment may, trigger / switch on, gene ; 6 / 7 two named examples ; ; - e.g. temperature and change in animal colour - high temperature and gender in crocodiles <u>UV</u> light and melanin production - wavelength of light and, flowering / fruit colour 8 environment effect usually greater on polygenes ; 9 environment may induce mutation (affecting phenotype) ; 10 AVP ;	7